



# WHARTON CONSULTING CLUB CASEBOOK 2022-2023

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KNOWLEDGE FOR ACTION  
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# **Case Interview Introduction**

# Case Interview Components

2-3 minutes

## Background, Setup, and Recap

- Active listening/taking thorough notes during prompt
- Summarize the problem
- Ask clarifying Q's
- Anticipate structure

4-5 minutes

## Framework Development & Explanation

- Structure thoughts in organized manner (<2 min)
- Give caser high level overview of buckets
- Drill down into each bucket
- Develop hypothesis & prioritize

3-5 minutes

## Brainstorming

- Probe for information
- Organize brainstorming into multiple buckets
- Be MECE - mutually exclusive, collectively exhaustive - with your ideas

8-10 minutes

## Math Exhibits & Analysis

- Provide overview of information
- Point out insightful details, utilize data from charts
- Note conclusions
- Walk through math problem steps BEFORE you calc

2-3 minutes

## Synthesis & Rec

- Drive case to conclusion
- Be answer FIRST, then provide supporting facts from case
- Take a definitive stance
- Utilize hard data from case
- Address risks & next steps

# Frameworks by case type

# Case Type 1: Profitability

## Overview

**Problem:** Client's earnings / profits has declined or stopped growing

**Objective:** Recommend ways to increase profits

## Sample Framework

### 1. Market

#### A. Industry

- Growth trends
- Market size
- Specific regulations

#### B. Competition

- Market shares
- Competitive advantage/weakness
- Entry barriers

### 2. Revenue

#### A. Price

- Pricing trends
- Compare to market

#### B. Volume

- Demand v Capacity
- Sales trends

#### C. Product Mix

- Demand v Capacity
- Sales trends

### 3. Cost

#### A. Fixed Costs

- PPE
- Overhead
- SG&A

#### B. Variable Costs

- COGS
- Labour
- Utilities

#### C. Supplier Power

#### D. Market Benchmark

### 4. Customer

#### A. Customer Segment

- Demographics
- Characteristics
- Preferences
- Willingness-to-pay

#### B. Channels

- Sales mix
- Operational efficiency
- Customer reach
- New opportunities

# Case Type 2: Market Entry

## Overview

**Problem:** Client is considering entering a new market

**Objective:** Recommend whether or not to enter (considerations: financial attractiveness, implementation success, risk assessment)

## Sample Framework

### 1. Market

#### A. Industry

- Growth trends
- Market size
- Specific regulations

#### B. Competition

- Competition intensity
- Competitor response
- Entry barriers

### 2. Financial

#### A. Current State

- Current profitability
- Financial capitals

#### B. New Market

- Capital investment
- Potential revenues
- Potential costs
- ROI on investment

### 3. Capabilities

#### A. Business Competency

- Market knowledge
- Comp advantages

#### B. Technical Competency

- Scaling operations
- Expansion management
- Other technical capabilities needed

### 4. Entry Strategy

#### A. Entry Methods

- Direct entry
- Acquisition
- Joint Venture

#### B. Considerations

- Market entry timing
- Piloting options
- Centralized vs Decentralized control
- Risks



# Case Type 3: Merger & Acquisition

## Overview

**Problem:** Client is considering acquiring another company

**Objective:** Evaluate the target company and recommend whether or not to pursue the deal

## Sample Framework

### 1. Market

**A. Are buyer and target in the same market?**

**B. Industry**

- Growth trends
- Market size
- Specific regulations

**C. Competition**

- Competition intensity
- Competitor response

### 2. Deal Evaluation

**A. Target's Financials**

- Current profitability
- Revenue growth
- Cost reduction

**B. Target's Competency**

- Management culture
- Technical capabilities
- Business expertise

**C. Deal ROI**

- Price vs Breakeven

### 3. Strategic Fit

**A. Acquisition Rationale**

- Vertical integration
- Horizontal integration
- New market entry

**B. Synergies**

- Cost driven
- Revenue driven
- Technical acquisition
- Response to competitor move

### 4. Risk Assessment

**A. Buyer capability**

- Prior acquisition experience
- Capital expenditure

**B. Post acquisition**

- Company cultural fit
- Integration process
- Organizational structure change

# Case Type 4: Non-Profit

## Overview

**Problem:** Client is a non-profit organization with a specific issue to explore

**Objective:** Understand the non-profit organization's core missions and find solutions to the specific problem presented

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## Sample Framework

### 1. Strategic Rationale

#### A. Mission of Non-Profit

- Poverty alleviation
- Education
- Health
- Environment

#### B. Stakeholders Interests

- Donors
- Beneficiaries
- Volunteers
- Paid staff

### 2. Solution Evaluation

#### A. Quantitative

- Cost of the project
- ROI on the project
- Benefits to stakeholders

#### B. Qualitative

- Alignment with non-profit mission
- Brand image of the non-profit

### 3. Capabilities

#### A. Financial

- Working capital
- Fundraising/donation

#### B. Operational

- Technical capabilities
- Labor force
- Required project expertise

#### C. Others

- Potential partnerships

### 4. Risk Assessment

#### A. Potential Challenges

- Government regulations
- Stakeholders buy-in
- Public reactions
- Media portrayal
- Other opposing organizations

# Case 1: Skedasky Farms

Case Type: Profitability; Market Entry  
Difficulty Level: Easy

# Skedasky Farms

## Problem Statement

Skedasky Farm is involved in the process of making, selling, and distributing white wines. In recent years, industry growth has stagnated. The CEO wants to increase revenue and profit, and has enlisted you to help to achieve this goal.

## Additional Information (Provide Upon Request)

- Skedasky grows all of their grapes
- Their farms are in California, with some processing facilities on-site, and another large facility located nearby
- Skedasky sells to wine distributors, and also sells directly to some stores in the state
- They sell a variety of grades of wine, from cheaper boxed wine to premium bottles
- The industry is very fragmented, and Skedasky has a comparable domestic market share to other major winemakers in its area

# Skedasky Farms

## Sample Framework

### 1. Current Revenue Streams

#### A. Increase Price

- Examine price elasticity
- Price by wine quality
- Price by wine type
- Single vs Bulk purchase

#### B. Increase Volume

- Increase share of shelf space
- Seek new distributors/channels
- Increase marketing efforts
- Introduce club membership

### 2. New Revenue Streams

#### A. New Products

- Red and sparkling wines
- Spirits
- Beer
- Food pairings
- Wine-related accessories

#### B. Expand Business Model

- Wine tours and tastings
- Weddings and events
- Sell grapes to other businesses

### 3. Cost Reductions

#### A. Fixed Costs

- Extend equipment's useful life
- Negotiate cheaper insurance
- Reduce SG&A costs
- Explore land opportunity costs

#### B. Variable Costs

- Use cheaper packaging
- Increase production efficiency
- Lower labour costs
- Reduce distribution costs

### Business insights

- “Most wineries in California do more than just sell wine. They provide wine tastings and tourist destinations, so there are more revenue opportunities in expanding to other service offerings.”
- “Given that Skedasky farms is currently grow their own grapes for white wines, I assume that it would be easy to also offer red wine, rosé, and sparkling wine by changing the production process slightly.”

# Skedasky Farms

## Q1: Increasing Price

Skedasky is considering changing the price of their mid level bottles of wine. Currently, 10,000 bottles are sold per year for \$20/each. Studies have shown that a 10% increase in the price will result in a 10% decrease in volume, and a 10% decrease in price will result in a 10% increase in volume. Which course should the company take to increase profit?

## Calculations

|        | Current   | 10% Price Increase | 10% Price Decrease |
|--------|-----------|--------------------|--------------------|
| Price  | \$20      | \$22               | \$18               |
| Cost   | \$10      | \$10               | \$10               |
| Margin | \$10      | \$12               | \$8                |
| Volume | 10,000    | 9,000              | 11,000             |
| Profit | \$100,000 | \$108,000          | \$88,000           |

### Tips:

Candidate needs to ask for profit margin. Profit margins are currently 50%, leading to \$10 cost/bottle

### Business Insights

An 8% increase in profit looks very attractive by simply increasing the price

# Skedasky Farms

## Q2: Diversifying into Beer

Skedasky thinks they can also start making and selling beer. What are some possible reasons this could be a good or bad idea for Skedasky?

### Advantages

#### Revenue Synergies

- Attract more customers
- Leverage existing brand equity
- Bundle sales for wine and beer
- Increase price for tasting/membership

#### Operational Synergies

- Economies of scale in operational logistics
- Some equipment can be used for both products
- Leverage existing distribution channels

### Disadvantages

#### Business Challenges

- Intense capital investment
- Brand equity dilution (Skedasky not known as a beer brand)
- Competitor response in both markets
- Lack of beer knowledge/expertise

#### Operational Challenges

- Setting up new production lines
- Creating new packaging for beer
- Increased marketing effort required

# Skedasky Farms

## Q3: Entering the Beer Market

If Skedasky were to enter the beer market, what are some ways they could enter? What are some pros and cons of each entry strategy?

|               |      |                                                                                                                                           |
|---------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Organically   | Pros | <ul style="list-style-type: none"><li>• More control</li><li>• No integration problems</li></ul>                                          |
|               | Cons | <ul style="list-style-type: none"><li>• Time and capital intensive</li><li>• Higher chance of failure due to lack of expertise</li></ul>  |
| Acquisition   | Pros | <ul style="list-style-type: none"><li>• Facilities and resources already exist</li><li>• Leverage brand name of acquired target</li></ul> |
|               | Cons | <ul style="list-style-type: none"><li>• Integration issues (culture clash)</li><li>• Acquisition cost could be expensive</li></ul>        |
| Joint Venture | Pros | <ul style="list-style-type: none"><li>• Shared risks and costs</li><li>• Gain new expertise and resources through partnership</li></ul>   |
|               | Cons | <ul style="list-style-type: none"><li>• Less control over the business</li><li>• Business and management culture clash</li></ul>          |



# Skedasky Farms

## Sample Recommendation

You are about to go into a meeting with the owner of Skedasky Farms. The client wants to know what recommendations you have for them.

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommendation</b> | <p><b>Immediate increase in profit: Increase the price of mid tier wine by 10%</b></p> <ul style="list-style-type: none"><li>• Research shows that profit could increase by roughly 8% in that category</li><li>• Help position brand in a more premium tier with price increase</li></ul> <p><b>Long term increase in profit: New product launch, but not beer</b></p> <ul style="list-style-type: none"><li>• Entering the beer market requires intense capital and time investment</li><li>• Selling beer could dilute client's brand image and reduce credibility on wine expertise</li></ul> |
| <b>Risks</b>          | <ul style="list-style-type: none"><li>• 8% profit increase in mid range wine might be relatively small for the entire company</li><li>• Not moving into beer market could limit profit gain in the long run</li></ul>                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Next Steps</b>     | <ul style="list-style-type: none"><li>• Conduct similar price elasticity of demand studies across existing wine products</li><li>• Research other potential product launch such as red and sparkling wine, etc.</li></ul>                                                                                                                                                                                                                                                                                                                                                                         |

## **Case 2: TissueCo**

Case Type: Profitability

Difficulty Level: Easy

# TissueCo - Prompt

## Problem Statement

Your client, TissueCo, is a large facial tissue manufacturer that sells to retailers. While the US facial tissue market suffered operational challenges due to Covid-19, the market is forecasted to recover and grow 5.8% CAGR in 2021. While TissueCo has seen some improved numbers as they adapt to the pandemic, their profitability margin has fallen below their main competitors Kelly-Clark and Proxy and Gumbo. The CEO has invited you to help them out.

## Additional Information (Provide Upon Request)

- The client is focused on US sales only
- TissueCo's profitability margin dropped below its competitors before the COVID-19 outbreak but has further worsened after COVID-19
- TissueCo produces tissues that are standard, extra soft, and tissues with lotion
- There is no specific goal to improve profitability

# TissueCo - Structure

## Sample Framework

### 1. Facial Tissue Market

#### A. Industry

- Growth trends
- Market size
- Recent changes in customer behavior / trends

#### B. Competition

- # of competitors and market share
- Profit margin / Cost structure

#### C. Regulatory environment

### 2. TissueCo

#### A. Product differentiation

- Product characteristics (e.g., soft, lotion, anti-viral)
- Packaging options (e.g., horizontal or vertical box, pocket)

#### B. Customer segment (e.g., at home, vs away from home)

#### C. Marketing strategy

#### D. Distribution strategy (e.g., supermarkets, convenience stores, online)

### 3. Financial Analysis

#### A. Revenue analysis

- Pricing strategy
- Volume dynamics
- Break-down by client groups and locations

#### B. Cost structure

- Fixed (e.g. rent, SG&A)
- Variable (e.g. labor, raw materials, utilities)

### 4. Growth Opportunities

#### A. Sales growth opportunities

- Lower prices
- New or improved characteristics
- Different packaging

#### B. Margin improvements

- Reduced volume per package
- Technology to reduce production costs

# TissueCo - Brainstorming

## Brainstorming prompt

What revenue growth ideas can you suggest?

# TissueCo - Brainstorming

## Sample Revenue Growth Ideas

### 1. Upgrade marketing strategy

#### A. Campaign

- Aggressive marketing campaign and promotions

#### B. Channels

- Expand to new social media channels to reach and cultivate younger audiences (a more germ-aware generation)

### 2. Change pricing (model)

#### A. Pricing

- Maintain price, but slightly slower volume per package
- Bundle packages together and/or offer bundles together with hand-sanitizer

#### B. Pricing model

- Subscription model (e.g., Amazon weekly)

### 3. Increase distribution efficiency

#### A. Distribution

- Online channels (online retailers and D2C)
- Volume dynamics

#### B. Partnerships

- Build partnerships with sectors affected by the rising concern of sickness (airlines, workspaces, schools)

### 4. Expand value proposition

#### A. New kinds of tissues

- Anti-viral
- More sustainable

#### B. New kinds of packaging

- The box could include a coupon for the next purchase

### Business insights

- Given low differentiation between facial tissues, high brand awareness and active marketing are likely key drivers of sales

# TissueCo - Market Sizing

## Market sizing prompt

The rise in the number of bacterial and viral infectious diseases, COVID-19 being top of mind, has led to the rise in demand for antiviral products. The client is considering launching a new product line to meet this demand, anti-viral tissues. How big do you think the US market in \$ dollars is for anti-viral facial tissues?

# TissueCo - Market Sizing

| Sample approach                                             |                                          |
|-------------------------------------------------------------|------------------------------------------|
| US Population                                               | ~320M people                             |
| % population that is very germ-conscious                    | 10%                                      |
| # of colds / flu per year per persons and duration          | ~2 colds per year that last 5 days       |
| # tissues used per day while sick                           | ~20 tissues                              |
| # tissues used per year for other reasons (e.g., allergies) | ~50 tissues                              |
| # tissue used per year                                      | $320M * 20\% * [(2*5*20)+50] \approx 8B$ |
| # tissues per box                                           | 100                                      |
| # of boxes                                                  | 80M                                      |
| \$ Avg price per box                                        | \$5                                      |
| \$ Total market                                             | <b>\$400M</b>                            |

## Provided information (if asked)

### Market sizing

- US population is 320M
- ~95% of the population uses tissues, so around 300M people
- Avg price for a box of tissues with 100 tissues is \$5

### Business insight:

- While representing a small portion of the market, germ-conscious consumers' WTP should be higher and could help improve profitability



# TissueCo - Math Prompt

## Math prompt

The rise in the number of bacterial and viral infectious diseases, COVID-19 being top of mind, has led to the rise in demand for antiviral product. The client is considering launching a new product line, anti-viral tissues. Will this increase their profitability?

## Additional Information (Provide Upon Request)

- Total sales in 2020 were \$3.1B (candidate *can* and *should* round to simplify calculations). Show Appendix 1 for revenue share and costs for other tissue segments. The candidate *should* calculate profit margins of all categories
- Anti-viral tissues are forecasted to generate \$150M in sales and \$30M in profit

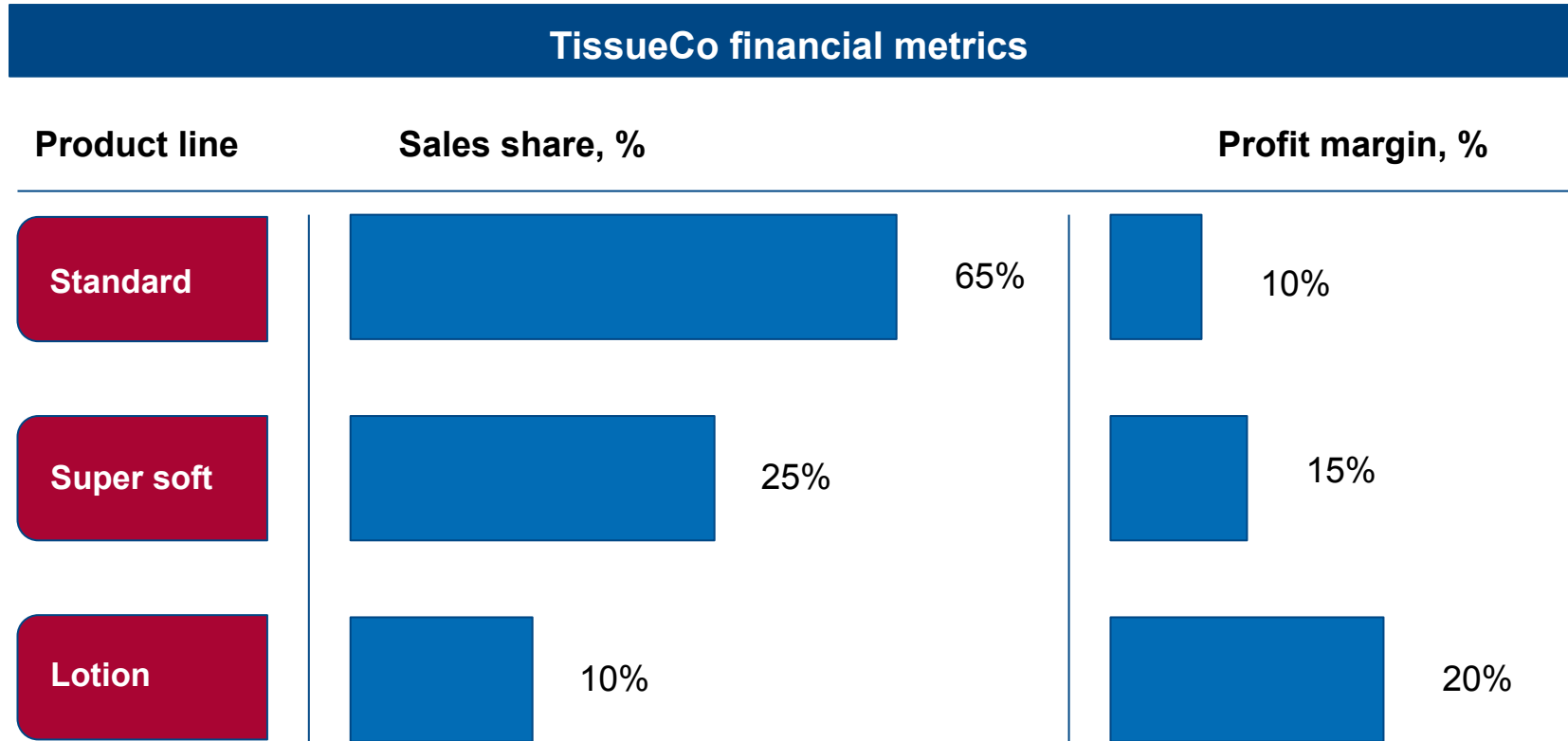
# TissueCo - Math Prompt

## Calculations

| Product    | Calculations                          | Profits (\$) / Profit Margin (%) |
|------------|---------------------------------------|----------------------------------|
| Standard   | $3B * 65\% * 10\% =$                  | \$195M                           |
| Super soft | $3B * 25\% * 15\% =$                  | \$112.5M                         |
| Lotion     | $3B * 10\% * 20\% =$                  | \$60M                            |
| Overall    | $195M + 112.5M + 60M = 367.5M / 3B =$ | $\approx 12\%$                   |
| Anti-viral | $30M / 150M =$                        | <b>20%</b>                       |

- The antiviral product line is forecasted to have a 20% profit margin, thus increasing the profitability of the company.

## Appendix 1. TissueCo financial metrics



# TissueCo - Recommendation

## Final Recommendation

What is your final recommendation?

## Sample Answer

**Recommendation:** In order to improve profitability, I recommend that the client launch the new antiviral tissues. 1) Demand for antiviral products has been increasing due to the recent pandemic; 2) Germ-conscious consumers' WTP is higher and we forecast a higher profitability margin of 20% compared to Standards 10% and Super Softs 15%.

### Risks:

- While we are forecasting profitability, competitive response could force us to lower prices.
- The current rise in demand due to the increased sensibility during the Pandemic could subside over the next few years.
- Clogs in the supply chain for antiviral products could limit production and increase COGS

### Next steps:

1. Evaluate the competitive landscape and build a sensitivity analysis based on different competitive responses
2. Take a deep dive into the market model to identify main growth drivers and determine which ones will be more sustainable
3. Map out the current Company's supply chain, identify possible weak points and offer potential workarounds.

# **Case 3: Big Yellow Bus**

Case Type: Market Entry

Difficulty Level: Easy

# Big Yellow Bus

## Problem Statement

Your client is a private equity fund considering the acquisition of the Big Yellow Bus Co, one of the leading manufacturers of school buses in the US. The client has engaged Bain to help determine whether or not to proceed with the investment.

## Additional Information (Provide Upon Request)

- BYB is the #1 player in the market by revenue, #2 by volume
- There are only 3 competitors in the market with relatively equal share. However, BYB was the clear leader 5 years ago
- BYB's price are 20% higher than its competitors
- The market has a fairly steady long-term 3% growth rate driven by GDP / population growth
- The customers are almost exclusively local cities and towns in the US

# Big Yellow Bus Co

## Sample Framework

### 1. School Bus Market

#### A. Industry

- Growth trends
- Market size
- Recent changes in customer behavior / trends

#### B. Competition

- # of competitors and market share
- Profit margin / Cost structure

#### C. Regulatory environment

### 2. Big Yellow Bus

#### A. Product differentiation

- Different kinds of buses
- Bus attributes vs competitors
- Extra services (e.g., maintenance, warranties)

#### B. Customer

#### C. Marketing strategy

- Salesforce and customer journey

#### D. Management expertise

### 3. Financial Analysis

#### A. Revenue analysis

- Pricing strategy
- Volume dynamics
- Break-down by client groups and locations

#### B. Cost structure

- Fixed (e.g. rent, SG&A)
- Variable (e.g. labor, raw materials, utilities)

### 4. PE Transaction

#### A. Price

- Financing
- Expected ROI
- Opportunity cost

#### B. Growth opportunities

- Value proposition
- Customer journey
- Marketing
- Expanding to new markets
- Additional services

### Business insights

- “In my experience, there isn’t a lot of differentiation between yellow school buses. As such, I would assume that the cost structure would drive the key competitive advantage.”
- “I would assume that our main customers would be price-sensitive school districts. Therefore, the pricing strategy is going to be an important factor in the growth or decline of market share.”

# Big Yellow Bus Co - Market Sizing

## Q1: Market Sizing

How large is the market for school buses in the US? (After the calculation: Do you believe this is an attractive market?)



# Big Yellow Bus Co - Market Sizing

| Sample approach                  |               |
|----------------------------------|---------------|
| US Population                    | ~320M people  |
| Life expectancy                  | ~80 years     |
| People per decade                | ~40M people   |
| School age commuters (~15 years) | ~60M children |
| % Taking buses                   | ~33%          |
| School-age children taking buses | ~20M children |
| Children per bus                 | ~50           |
| Buses required                   | 400K buses    |
| Average life of buses            | 10 years      |
| Buses sold per year              | 40k           |

## Provided information (if asked)

### Market sizing

- ~33% of schoolchildren take buses to school
- ~50 students ride each buses

### Attractive market

- Market growth is traditionally GDP / population at 3% per year; last two years have seen 6% growth
- Pull-through demand ahead of changing emissions regulation

### Business insight:

- “This seems to be a good size market with consistent growth. Still, in order to evaluate BYB, do we have information of market share trends?”

# Big Yellow Bus Co - Chart Analysis

## Q2: Competitive landscape

Take a look at Exhibit #1, what does the competitive landscape data imply?

### Sample Answer

There are **two** major takeaways:

- BYB's share of revenue is higher than its share of volume
  - Why? Its prices are higher than peers by 20%. This is important because its a commodity product and customers are highly price-sensitive.
  - 5 years ago, BYB's market share was at 60%.
- BYB's cost structure is high relative to peers
  - The gross margin is a big differentiation and BYB needs to focus on reducing its COGS
  - Some possible hypotheses:
    - Materials, equipment, and labors will probably be most significant cost drivers. Competitors could be more successful in outsourcing or achieving economies of scale
    - Competitors A & B could be a part of larger industrial trucking firms where there are synergies

# Big Yellow Bus Co - Exhibit #1

| Competitive Landscape        |                   |              |              |
|------------------------------|-------------------|--------------|--------------|
|                              | Big Yellow Bus Co | Competitor A | Competitor B |
| <b>Market share analysis</b> |                   |              |              |
| Market share (\$)            | 38%               | 34%          | 28%          |
| Market share (units)         | 34%               | 36%          | 30%          |
|                              |                   |              |              |
| <b>Margin analysis</b>       |                   |              |              |
| Revenue                      | 100%              | 100%         | 100%         |
| Gross margin                 | 25%               | 35%          | 36%          |
| Operating margin             | 15%               | 25%          | 26%          |

# Big Yellow Bus Co - Recommendation

## Final Recommendation

What is your final recommendation?

## Sample Answer

**Recommendation:** I recommend that the PE fund *not* acquire Big Yellow Bus Company.

Reason #1: Even if pricing of transaction is attractive, BYB is in a difficult competitive position. BYB is losing share to two lower-cost competitors with significant built-in cost advantages given their ownership structures.

Reason #2: Further, cost is the biggest driver in the market as buses become increasingly commoditized and local cities & towns become more price sensitive in the face of budget deficits.

Conclusion: Overall, not an attractive investment.

### Risks:

Economic decline / immigration patterns could accelerate demand beyond 3% growth, but even in this case BYB is unattractive.

### Next steps:

Make sure PE fund does not own-portfolio companies that might allow for procurement synergies

# Case 4: Penn and Teller

Case Type: Profitability  
Difficulty Level: Medium

# Penn and Teller

## Problem Statement

Our client is Caesars, one of the largest gaming and resort companies in the world. Penn and Teller have been the headlining show at the Rio Hotel since 2001, and they are currently the longest-running headlining show in Vegas history. Caesars, who owns the Rio Hotel where the duo performs in the 1,500 seat P&T Theater, is wondering if their act has become stale. Penn & Teller's annual contract is about to expire, and before Caesars has a meeting with the duo, they have asked our company's advice on whether or not to re-sign the act for another year.

## Additional Information (Provide Upon Request)

**What goal does Caesars have?** As one of the largest entertainment companies on the Strip and around the world, Caesars solely cares about the company's bottom line profitability.

**How long is a contract for Penn & Teller?** As is the Vegas standard, all contracts are for a one-year time period.

**Why does Caesars think the show is stale?** The show has plateaued with no change in bottom line profitability in the past five years.

**Who are Caesars'/P&T's biggest competitors?** Caesars' biggest competitor is MGM Resorts, which also operates about 33% of the Strip. P&T compete against a whole bevy of nighttime entertainment, such as other shows, nightclubs, and gaming.

# Penn and Teller

## Sample Framework

### 1. Revenue from P&T show

#### A. Price

- Types of tickets
- Price of tickets

#### B. Volume

- Capacity of the theatre
- Sell through of said capacity by ticket type

### 2. Costs

#### A. Variable

- Consumables (props, handbills, etc.)
- Labor (cleaning, ushers, stagehands, etc.)
- Utilities (electricity, etc.)

#### B. Fixed

- Marketing
- SG&A
- Annual contracts
- Insurance

### 3. Impact on other revenue

- Room fees
- Gambling
- Dining

### 4. Alternatives

- Replacement acts and cost/benefit
- Alternate use of the space (more casino space, buffet, etc.)

### Business insights

- “Beyond just the direct P&L impact to Caesars, I also want to understand what other benefits the show provides. Good word of mouth, additional revenue etc.”

# Penn and Teller - Profit (Revenue)

## Q1: Revenue

**How profitable is the Penn and Teller show? Only provide the information below when asked.**

**How many seats?** The theater has 1500 seats.

**Are all the seats the same?** No, there are three categories, A, B, and C. There are 300 Category A seats (best seats in the house), 800 Category B seats, and 400 Category C seats (balcony, or in the back of the theater).

**Do the seats always sell out?** No. On average, 100% of the A, 80% of the B, and 50% of the C are sold.

**What are the prices?** Category A costs \$120, B costs \$75, C costs \$55.

**How many shows do P&T perform?** They perform at Vegas' standard 6 shows per week, 40 weeks per year (they have 1 day off per week and three months per year when they work on other endeavors like TV appearances and book deals).

**How has revenue changed?** These numbers have been static over the past 5+ years, hence why Caesars thinks the show has become stale.



## Penn and Teller - Revenue Calculations

| Calculation             |                        |                       |                            |                         |                       |
|-------------------------|------------------------|-----------------------|----------------------------|-------------------------|-----------------------|
| <u>Category</u>         | <u>Number of Seats</u> | <u>% Sold</u>         | <u>Attendance per show</u> | <u>Price per Ticket</u> | <u>Ticket Revenue</u> |
| CAT A                   | 300                    | 100%                  | 300                        | \$120.00                | \$36,000              |
| CAT B                   | 800                    | 80%                   | 640                        | \$75.00                 | \$48,000              |
| CAT C                   | 400                    | 50%                   | 200                        | \$55.00                 | \$11,000              |
|                         |                        |                       |                            |                         |                       |
| <u>Revenue per Show</u> | <u>Shows per Week</u>  | <u>Weeks per Year</u> | <u>Revenue per Year</u>    |                         |                       |
| \$95,000                | 6                      | 40                    | \$22.8M                    |                         |                       |

# Penn and Teller - Profit (Cost)

## Q1: Cost

**What are the costs of the Penn and Teller show? Only provide the information below when asked.**

**What are the costs to putting on the show?** Costs are broken into six main buckets

- 1) Both Penn and Teller each make \$2,000,000 **per year**
- 2) The Crew (showgirls, ushers) in total costs \$2,000 **per show**
- 3) Housekeeping costs \$1,000 **per week**
- 4) The Props (the doves, playing cards, etc) costs \$200 **per show**
- 5) Utilities(the lights) cost \$52,000 **per year**
- 6) SG&A ( box office staff, etc) costs \$15,000 **per month**

**Does housekeeping clean year round?** No, they are only needed the 40 weeks the show is live

**Is SG&A paid year round?** Yes, that is a 12 month per year expense

**How have costs changed?** These costs have been static over the past 5 years

## Penn and Teller - Cost Calculations

| Calculation                 |                      |                    |
|-----------------------------|----------------------|--------------------|
| Expense                     | Formula              | Total              |
| Penn and Teller (2 of them) | $\$2,000,000 * 2 =$  | \$4,000,000        |
| Crew (6 shows, 40 weeks)    | $\$2,000 * 6 * 40 =$ | \$480,000          |
| Housekeeping (40 weeks)     | $\$1,000 * 40 =$     | \$40,000           |
| Props (6 shows, 40 weeks)   | $\$200 * 6 * 40 =$   | \$48,000           |
| Utilities (1 year)          | $\$52,000 * 1 =$     | \$52,000           |
| SG&A (12 months)            | $\$15,000 * 12 =$    | \$180,000          |
| <b>Total Cost per Year</b>  |                      | <b>\$4,800,000</b> |

## Penn and Teller - Profit and Margin

| Calculation                    |                     |
|--------------------------------|---------------------|
| Line Item                      | Amount              |
| Revenue                        | \$22,800,000        |
| Cost                           | \$4,800,00          |
| <b>Profit</b>                  | <b>\$18,000,000</b> |
| <b>Margin (Profit/Revenue)</b> | <b>~80%</b>         |

# Penn and Teller - Brainstorm

## Question

What can Caesars do to make the current show more profitable?

## Sample Answer

### Revenue

- Ticket Prices
  - Increase prices?
- Quantity Sold
  - Switch some B seats into A's, C's into B's (higher margin seats)
  - Offer bundle discounts
  - Put on more shows
- Other
  - Retool concession stand
  - Add gift shop with better Souvenirs
  - Shorten the show so viewers gamble more (higher margin activity)
  - Bundle tickets with dinner or other activities (like an overnight stay)

### Costs

- Revisit contract with P&T
- Check union vs non union labor

# Penn and Teller - Question

## Question

Caesars has the opportunity to switch the P&T show for one featuring the recent winner of America's Got Talent. Should they?

## Clarifying Questions and Answers

**What goal does Caesars have?** As one of the largest entertainment companies on the Strip and around the world, Caesars solely cares about the company's total bottom line profitability

**How long is a contract for the AGT?** As is the Vegas standard, all contracts are for a one year time period

**Are there any costs to making the switch?** Putting in a new show will require a reconfiguration of the theater and a one time marketing blitz This would cost \$24,000,000 to do and would occur before the first show

**What are the revenues of the new show?** The new show would bring in only \$90,000 in ticket sales per show (compared to P&T of 95 All other sources of revenue ( concessions, etc are insignificant)

**What are the costs of the new show?** The new show would cost significantly less than the P&T, costing 3 M per year as opposed to \$48,000,000

# Penn and Teller - Profit for New Act

## Change in Profits

| What happened                        | Formula                          | Total                           |
|--------------------------------------|----------------------------------|---------------------------------|
| Less Revenue per show (-\$5,000)     | $-\$5,000 * 240 \text{ shows} =$ | $-\$1,200,000 \text{ per year}$ |
| Lower show costs per year            | $\$1,800,000 \text{ per year}$   | $\$1,800,000 \text{ per year}$  |
| New Annual Profit                    | $-\$1,200,000 - \$1,800,000 =$   | $\$600,000 \text{ per year}$    |
| Payback Period (\$2,400,000 upfront) | $\$2,400,000 / \$600,000 =$      | 4 years                         |

## Sample Answers

### YES:

Penn & Teller is stale, and the new America's Got Talent act is fresh

America's Got Talent is \$600,000 more profitable each year

### NO:

4 year is a long time in such a competitive market

America's Got Talent winner might not have long - lasting appeal

# Penn and Teller - Recommendation (Resign Penn & Teller)

## Final Recommendation

The SVP of Entertainment wants to know what we discussed. What recommendation do you have for Caesars?

## Sample Answer

|                       |                                                                                                                                                                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommendation</b> | I recommend that Caesar's resign Penn and Teller. The show remains very profitable generating \$18,000,000 profit on \$22,800,000 of revenues. The current America's Got Talent Alternative is more profitable, however it will take 4 years to recoup the initial investment. |
| <b>Risks</b>          | <ul style="list-style-type: none"><li>● Assumption that Penn and Teller will maintain their popularity</li><li>● Assuming that costs will not continue to rise (e.g. Penn and Teller wanting more)</li></ul>                                                                   |
| <b>Next Steps</b>     | <ul style="list-style-type: none"><li>● Moving forward, we can look at how to further increase revenues by doing XXXX</li><li>● Or minimize costs by doing XXXX and further look into XXXX</li></ul>                                                                           |



# Penn and Teller - Recommendation (Do not resign Penn & Teller)

## Final Recommendation

The SVP of Entertainment wants to know what we discussed. What recommendation do you have for Caesars?

## Sample Answer

|                       |                                                                                                                                                                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommendation</b> | I recommend that Caesar's does not resign Penn and Teller. The show has become stagnant with no growth, and while still profitable, the new America's Got Talent show is fresher and would bring in an additional \$600,000 in profits.             |
| <b>Risks</b>          | <ul style="list-style-type: none"><li>• 4 year payback period for the initial upfront costs is long in a competitive market</li><li>• No guarantee that the America's Got Talent has lasting popularity</li></ul>                                   |
| <b>Next Steps</b>     | <ul style="list-style-type: none"><li>• Conduct additional market research to see how popular and resilient an America's Got Talent act would be</li><li>• See if there are alternatives that more profitable with lower payback periods.</li></ul> |

# Case 5: Franchising Gyms

Case Type: Profitability; Market Entry  
Difficulty Level: Easy

# Franchising Gyms

## Problem Statement

Your client is a gym franchisor. The client focuses on small gyms (average of 3000 square feet) with standard fitness equipment, but no group fitness classes. The gyms are typically located in local strip malls. They are open 24 hours per day and 7 days a week. Members enter the gym via an access card and staffing at the gym is minimal. The business is growing rapidly. Our client has asked for your help to understand where to grow their business and how to improve profitability in this highly competitive market.

## Additional Information (Provide Upon Request)

- Our client's primary goal is to grow their business and improve profitability
- Gym membership is a flat monthly fee
- Our client has franchisees across the U.S.

# Franchising Gyms

## Sample Framework

### 1. Market Opportunity

#### A. Market size and growth

- Size of the market
- Growth expectations
- Current and upcoming trends in the market

#### B. Competition in the market

- Market concentration/fragmentation
- Ease of market entry/barriers
- Who are the key competitors
- Key differentiating factors

### 2. Financial Analysis

#### A. Revenue Streams

- Monthly membership fees
- Personal training
- Merchandise (if any)
- Vending machines/snacks

#### B. Cost Structure

- *Fixed Costs*: space rental, fitness equipment, franchise fees, SG&A
- *Variable Costs*: utilities, labor, maintenance

### 3. Target Customers

#### A. Customer segmentation

- Demographic segmentation
- Geographic segmentation
- Behavioral segmentation
- Socioeconomic segmentation

#### B. Customer preferences

- Willingness to pay
- Basic vs upscale gyms
- Additional services and product offerings

### Business insights

- Since client's gym only offers standard fitness equipments, they are targeting customers who want lower price and basic fitness products and services
- Client's gym is only very minimal in terms of staffing and equipments that cutting cost could be more challenging here. The biggest cost contribution here is probably rent

# Franchising Gyms

## Q1: Customer Segment Analysis

Our client wants to determine which geographic location to focus their growth efforts. Based on the customer segment analysis presented (see exhibit on the next page), what advice do you have for the client?

### Sample Response

|          | Costs for Franchisee | Gym Space Location Availability | Convenience for Members | Segment Fitness Interest | Segment Spare Income | Competition |
|----------|----------------------|---------------------------------|-------------------------|--------------------------|----------------------|-------------|
| Urban    | High (☹)             | Low (☹)                         | High (☺)                | High (☺)                 | Medium               | High (☹)    |
| Suburban | Medium               | Medium                          | Medium                  | Medium                   | Medium               | Medium      |
| Rural    | Low (☺)              | High (☺)                        | Low (☹)                 | Medium                   | Medium               | Low (☺)     |

#### Key Takeaways:

- Rural segment is very attractive because of the lower cost, high availability of gym space, and low competition
- Even though convenience for members is low in rural areas, the customers still have relatively high fitness interest which is favorable for our client
- Given our client's current no-frills business model, a low cost and low competition market is the best option for growth

# Franchising Gyms

## Exhibit - Customer Segment Analysis

|          | <b>Costs for Franchisee</b> | <b>Gym Space Location Availability</b> | <b>Convenience for Members</b> | <b>Segment Fitness Interest</b> | <b>Segment Spare Income</b> | <b>Competition</b> |
|----------|-----------------------------|----------------------------------------|--------------------------------|---------------------------------|-----------------------------|--------------------|
| Urban    | High (☹)                    | Low (☹)                                | High (☺)                       | High (☺)                        | Medium                      | High (☹)           |
| Suburban | Medium                      | Medium                                 | Medium                         | Medium                          | Medium                      | Medium             |
| Rural    | Low (☺)                     | High (☺)                               | Low (☹)                        | Medium                          | Medium                      | Low (☺)            |

# Franchising Gyms

## Q2: How many more members needed?

Our client is currently not making any profit. If our client wants to increase profit by 10%, how many more members on average does a location need to acquire? (see current revenue and cost breakdown on the next page)

### Current Revenue and Cost Breakdown Per Location

| Revenue Per Member      |              |
|-------------------------|--------------|
| Monthly Membership Dues | \$40 / month |
| Vending Machine         | \$5 / month  |
| Personal Training       | \$15 / month |

| Cost           |                                                                 |
|----------------|-----------------------------------------------------------------|
| Rent           | \$2 / sq ft / month                                             |
| Equipment      | \$144k initial cost<br>(3 year depreciation, \$0 salvage value) |
| Labor          | \$20 / hour (2 people on staff at a time)                       |
| Utilities      | \$400 / month                                                   |
| Franchise Fees | \$300 / month                                                   |
| Marketing      | \$100 / month                                                   |

# Franchising Gyms

## Calculations

| Revenue Total: \$60/month |                  |
|---------------------------|------------------|
| Per Member                | \$60 / month     |
| Cost Total: \$37,680      |                  |
| Rent                      | \$6,000 / month  |
| Equipment                 | \$4,000 / month  |
| Labor                     | \$26,880 / month |
| Others                    | \$800 / month    |

|                                     |                            |
|-------------------------------------|----------------------------|
| Current # of Members                | $37680 / 60 = 628$ Members |
| Increase Profit by 10%, New Revenue | \$41,448 / month           |
| New # of Members                    | ~ 690 Members              |
| Additional Members Needed           | 29 Members                 |

### Key Takeaways:

- Additional 62 members is about 10% of current number of members
- These numbers seem achievable, especially if the client targets rural areas where there is still relatively strong interest in fitness with low competition
- Client should explore ways to attract more customers to understand how achievable it is to obtain 62 additional members



# Franchising Gyms

## Q3: Revenue Growth Opportunities

What are some potential revenue growth opportunities the client can pursue?

### Sample Response

#### 1. Price

- Increase current membership fees
- Utilize tiered membership based on accessibility
- Consider multi-period discounted pricing
- Add initiation fee for the first month

#### 2. Quantity

- Increase # of gyms franchised
- Offer family plans to gain more members
- Partner with corporate companies to offer discounted memberships
- Use referral bonuses to acquire more members

#### 3. Product/Service

- Juice and snack bar
- Selling merchandise (clothing, weights, yoga mats, etc.)
- Adding group workout classes or other training services
- Offer day pass options for short term gym usage

# Franchising Gyms

## Sample Recommendation

Please provide a recommendation for our client

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommendation</b> | <ol style="list-style-type: none"><li>1. Based on our client's current business model, it makes more sense to focus on expanding franchising effort in rural markets where cost and competition are low</li><li>2. Our client could improve profitability by 10% if they can increase membership by 62 members per gym which seems to be feasible goal</li><li>3. Our client could explore attracting more members by adding new product/service offerings such as new workout classes, or by offering more attractive pricing plans</li></ol> |
| <b>Risks</b>          | <ul style="list-style-type: none"><li>• Adding new services/products might make client's gym less distinct from competitors</li><li>• Target audience might not want the new services/product offerings</li></ul>                                                                                                                                                                                                                                                                                                                              |
| <b>Next Steps</b>     | <ul style="list-style-type: none"><li>• Conduct a customer preference survey to understand what new services/products best align with target customer interests</li><li>• Identify best rural market locations to add new franchise locations</li></ul>                                                                                                                                                                                                                                                                                        |

# Case 6: CarCo Autoparts

Case Type: Profitability

Difficulty Level: Easy

# CarCo Autoparts

## Problem Statement

Your client is CarCo, a mechanics and auto parts company in the MidWest. CarCo provides automotive services to customers (tires, oil changes, engine repair, etc.). The business is family owned and operated, and has historically generated \$25 million in profits annually. Recently, the business has experienced far lower profits than usual, namely \$10 million. While the business is still profitable, a similar drop in profits would be catastrophic. You have been hired to assess the root cause of decreased profits and help turn the business around.

## Additional Information (Provide Upon Request)

- CarCo only operates in the MidWest and candidate can assume it's a single location
- CarCo mainly offers three products/services: regular tires, oil change, and engine overhaul
- The profit decline is only observed in the last few years
- No specific financial target or timeline. CarCo just want to identify root cause and find solutions to these problems

# CarCo Autoparts

## Sample Framework

### 1. Market issues (Not specific to any company)

- A. Overall market for mechanic services is declining due to renewable energy and fewer vehicles in the market**
- B. Number of cars left unchanged, but cars need fewer repairs than they used to, thereby shrinking market**
- C. People fix cars themselves**

### 2. Revenue Increase for CarCo

- A. Prices have fallen**
  - Willingness to pay has fallen
  - Price war with competitors
  - Economic hardship
- B. Quantity sold has dropped** (for reasons apart from market issues)
  - Competitors stole market share
    - Client's quality has deteriorated
    - Turnaround time has increased, etc.

### 3. Cost Increase for CarCo

- A. Fixed costs**
  - Garage/facility rental
  - Utilities
  - Equipment
  - SG&A
- B. Variable costs**
  - Labor costs - recruiting / training
  - Particular car parts (software?)
  - Timing costs for lengthy repairs

### Business insights

- A very good answer would hypothesize regarding the root cause of declining profitability. Ex: I doubt a mechanic's costs would be in the \$15 million range, thus we should focus more on revenue and/or market factors
- A very good answer leverages personal experience regarding what is happening in the world when commenting on the automotive industry (Ex. shift to electric cars)

# CarCo Autoparts

## Q1: Identify Revenue Streams

The company's cost structure has remained constant, and there are no external market factors that are relevant here. What are the revenue streams you would expect this mechanic to have?

### Sample Response

#### 1. Product

- Hardware - tires, engines, brakes, alternators
- Software - speakers, radios, touchscreens, chargers
- Miscellaneous - oil, coolant

#### 2. Services

- Yearly inspections
- Winterizing, oil change
- Post-accident repair
- Scrapping the car

#### 3. Passive Income

- Investments
- Property

# CarCo Autoparts

## Q2: Product/Service Mix

CarCo does sell a mix of products and services to customers. Historically the total quantity of products/services sold has remained constant at 1 million. In addition the prices of these offerings have not changed at all according to the chart below. Can you explain what has happened, and how you would suggest the client to increase profits?

| Product         | Price | Profit Margin (@50%) |
|-----------------|-------|----------------------|
| Regular Tire    | \$10  | \$5                  |
| Oil Change      | \$30  | \$15                 |
| Engine Overhaul | \$200 | \$100                |

*\*\*Margins are 50% across all product/service lines*

# CarCo Autoparts

## Sample Response: Cause of Profit Decline

Despite a constant total number of products sold, customers have purchased proportionally more tires and oil change and proportionally fewer engine overhauls. Thus, the declining profitability is due to a switch from higher-margin items to lower-margin items.

## Sample Response: Revenue Opportunities

|    |                                                                                                                                                                 |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Increase customer base or expand number of shops - may be tough if population is roughly constant                                                               |
| 2. | Increase product offering to incentivize customers to buy more; bundled packages to increase spend                                                              |
| 3. | Reduce the price of engines, as current profits are \$100 and our competitors may have undercut us                                                              |
| 4. | If the market is moving towards “do it yourself”, CarCo could sell engines without the service component; also expand into selling DIY kits in product category |
| 5. | Marketing campaign in the community, or other promotions                                                                                                        |



# CarCo Autoparts

## Q3: Strategy Risks

CarCo approaches you and is convinced the right strategy moving forward is adding Star Tires to its product line. These tires would sell for \$25 with a profit margin of 60%. What are the risks of selling star tires?

### Sample Response

|                 |                                                                                                                                                                                    |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cannibalization | People who buy star tires will buy them instead of regular tires                                                                                                                   |
| Customers       | Will our customers want to buy more expensive tires? They are already shying away from our more expensive products. Are star tire shoppers our current customers or new customers? |
| Brand           | As a family-owned business, would selling star tires force us to increase the service quality and become a more premium brand? Would we lose business this way?                    |
| Permanence      | Are star tires a fad? If they last much longer than regular tires, will they need to be replaced less often and require less maintenance?                                          |
| Execution       | Do we have a stable supplier, and do we have the inventory to support this? Can our mechanics change these tires?                                                                  |

# CarCo Autoparts

## Sample Recommendation

Please provide a recommendation for our client

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommendation</b> | <ol style="list-style-type: none"><li>1. We discovered profits have fallen 60% due to drop of engine repairs being sold. While our other products have sold better, the increase has been insufficient to compensate the loss from engines</li><li>2. The mechanic could change the pricing of engine overhauls to better compete in the market - we would want to see what other mechanics charge</li><li>3. CarCo could also explore adding on star tires since it is expected the profit margin will be 60% (higher than current product / service line)</li></ol> |
| <b>Risks</b>          | <ul style="list-style-type: none"><li>• Lowering engine overhaul price might further reduce profits if there is limited demand in general</li><li>• Cannibalization could take customers away from regular tires</li></ul>                                                                                                                                                                                                                                                                                                                                            |
| <b>Next Step</b>      | <ul style="list-style-type: none"><li>• Further analyze the market demand and competitor pricing for engineer overhauls</li><li>• Conduct in depth cost-benefit analysis of adding star tires before pursuing the opportunity</li></ul>                                                                                                                                                                                                                                                                                                                               |

# **Case 7: Dr. Kelso Ophthalmology Practice**

Case Type: M&A

Difficulty Level: Medium to Hard

# Dr. Kelso Ophthalmology Practice

## Problem Statement

Your client, a PE firm, is considering an investment in Dr. Kelso's ophthalmology practice, located outside Philadelphia. Should they make the investment?

## Additional Information (Provide Upon Request)

- How does Dr. Kelso's business make money? What is ophthalmology?
  - An ophthalmologist is an eye doctor
  - Dr. Kelso's practice provides a full range of ophthalmology services including exams, surgeries, and prescribing and administration pharmaceuticals
  - Accepts major commercial and government insurance
- What is the ownership structure today?
  - Today, Dr. Kelso and 3 other doctor partners are equity owners in the business
  - They are only paid their share of the profits. They do not take a salary
- What are the terms of the deal?
  - Dr. Kelso's Practice is requesting \$15M for a 50% stake in the business
- What are the PE firm's objectives?
  - To make a good ROI on the investment

# Dr. Kelso Ophthalmology Practice

## Sample Framework

| 1. Market                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2. Financials                                                                                                                                                                                                                                                                                                                                                                   | 3. Opportunities                                                                                                                                                                                                                                                                                                                                                                                                  | 4. Deal Terms                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A. Market Size</b> <ul style="list-style-type: none"><li>Number of customers (Insurance vs cash)</li><li>Market growth</li><li>Demand</li></ul> <b>B. Competitors</b> <ul style="list-style-type: none"><li>Number of other competitors</li><li>Hospital owned vs independent</li><li>Focus - full suite vs focused</li><li>Affiliation with referrers</li></ul> <b>C. Trends</b> <ul style="list-style-type: none"><li>Teleconsultation</li><li>Growth of cosmetic surgeries like LASIK</li></ul> | <b>A. Revenue</b> <ul style="list-style-type: none"><li>Price (avg service fee, collections rate, insurance vs cash price)</li><li>Volume (number of patients, insurance vs cash mix, service mix)</li></ul> <b>B. Cost</b> <ul style="list-style-type: none"><li>Variable (labor, materials, marketing etc.)</li><li>Fixed (rent, loans, equipment, utilities, etc.)</li></ul> | <b>A. Revenue Synergies</b> <ul style="list-style-type: none"><li>Increase reimbursement rates via greater market power</li><li>Increase volume via combination with referrers</li><li>Increase collection rate via better business Management</li></ul> <b>B. Cost Synergies</b> <ul style="list-style-type: none"><li>Decrease practice management costs through scale with other assets in portfolio</li></ul> | <b>A. ROI/Payback Period/NPV</b><br><b>B. Ownership structure (PE &gt; 50%?)</b><br><b>C. Why are doctors selling?</b><br><b>D. Risks</b><br><b>E. Other Opportunities</b> |

### Interviewer guidance

- After framework, if needed, guide the candidate to focus on the profitability of the company (Market would also be a reasonable place to go but is not the focus of this case).

# Dr. Kelso Ophthalmology Practice

## Exhibit 1: Financial Forecast

The PE firm prepared a financial projection for Dr. Kelso's practice for next year before the deal takes place. As a first step, what is the implied valuation of the company today?

| Product                 | 2022F       |
|-------------------------|-------------|
| Revenue                 | \$20M       |
| Costs                   | \$16M       |
| <b>Operating Profit</b> | <b>\$4M</b> |
| Taxes (20%)             |             |
| <b>Net Profit</b>       |             |

### Assumptions:

WACC = 13%  
Growth Rate = 3%

*Note: Dr. Kelso's has no depreciation, debt, or investment in CapEx*

# Dr. Kelso Ophthalmology Practice

## Exhibit 1: Answer Key

| Product                 | 2022F  |
|-------------------------|--------|
| Revenue                 | \$20M  |
| Costs                   | \$16M  |
| <b>Operating Profit</b> | \$4M   |
| Taxes (20%)             | \$0.8M |
| <b>Net Profit*</b>      | \$3.2M |

### Notes:

$\text{Value} = \text{FCF} / (\text{WACC}-g)$

Valuation = 32M PE stake at  
50% is worth \$16M.

ROI is \$16M/\$15M or ~7%  
(not good)

*\*Note that net profit is equivalent to FCF because we assume no depreciation or capex*

# Dr. Kelso Ophthalmology Practice

## Question 2

If the PE firm acquired a stake in Dr. Kelso's practice, how could they increase the value of the company?

## Brainstorm

Candidate should brainstorm revenue drivers, cost drivers, and synergies (see sample framework for examples)



# Dr. Kelso Ophthalmology Practice

## Question 3

After initial research the PE firm believes they can improve revenue by 5% via improved pricing and collections. In addition, through cost synergies, they believe they can reduce practice management costs by \$1M. What would the PE client's stake be worth with these changes?

## Answer

|                  |                                                                                                                                                                                                                                     |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Revenue          | Revenue = $1.05 * \$20M = \$21M$ ; Costs = $\$16M - \$1M = \$15M$                                                                                                                                                                   |
| Operation Profit | Operation Profit = $\$6M$ ; Taxes = $\$1.2M$                                                                                                                                                                                        |
| Net Profit       | $\$4.8M$ ; Valuation = $\$48M$ . PE stake = $\$24M$                                                                                                                                                                                 |
| Takeaway         | PE firm would make a 60% ROI ( $\$24M$ value / $\$15M$ investment) -> a good investment                                                                                                                                             |
| Bonus            | An outstanding candidate would say, if volume declines by ~10% from $\$21M$ in revenue to $\$19M$ in revenue, then the value of the business declines to just $\$36M$ and the PE firm would just barely breakeven on its investment |

# Dr. Kelso Ophthalmology Practice

## Question 4

How would compensation for Dr. Kelso and his 3 partners change after the deal? How do you think this would impact volume?

## Answer

|                                  |                                                                                                                                                                              |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Current Setup                    | Dr. Kelso and partners receive no salary; compensation today (pre-deal) = Net profits of the business, \$3.2M                                                                |
| New Setup                        | After the deal, with \$4.8M net profit, the partners receive a \$24M payout and 50% of the profit stream, or \$2.4M per year                                                 |
| How does this affect motivation? | As 50% equity owners, the doctors have less incentive to work hard. They only get 50% of incremental profits from their efforts                                              |
| How will this affect volume?     | Given this incentive change, we would expect volume to go down, and after receiving a \$24M payout, even less incentive to work hard (They'll have new ski houses to enjoy!) |
| Takeaway                         | Given the reduced profit, doctors' incentive decreases, which drives down volume, and reduction of volume leads to the value of the business decreasing.                     |

# Dr. Kelso Ophthalmology Practice

## Question 5

How can the deal be changed to make sure we keep current volume growth at 3%?

## Sample Answer

Incentive-based salaries / bonus tied to volume, contractual requirement to maintain strong effort, vest the payout over multiple years and condition on volume, inspire the doctors to consider the patient impact, hire new doctors/other staff to improve leverage per doctor

## Question 7

The client determines that they need to pay \$0.6M in performance-based compensation to Dr. Kelso and his partners to keep them sufficiently incentivized and continue to grow volume. With this change, what would the value of the PE firm's stake be?

## Answer

Operating profit = \$5.4M, or \$0.6M lower  
Taxes = \$1.08M; Net Profit = \$4.3M (ok to round); Valuation = \$43M; PE stake = 21.5M; ROI = 43%

# Dr. Kelso Ophthalmology Practice

## Recommendation

Candidate can make an argument either way.

|                         |                                                                                                                                                                                                                                                                        |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sample Arguments</b> | <ul style="list-style-type: none"><li>• Argument to invest: 1) Strong ROI 2) grow revenue by 10%: increase pricing power / collections rate, 3) reduce practice management costs by \$2M, 4) good ways to de-risk effort dis-incentives, keeping good volume</li></ul> |
| <b>Sample Risks</b>     | <ul style="list-style-type: none"><li>• Doctors will not have the proper incentive to continue working as hard</li></ul>                                                                                                                                               |
| <b>Sample Next Step</b> | <ul style="list-style-type: none"><li>• Explore which incentive structures will be the most effective and is agreeable to the doctors</li></ul>                                                                                                                        |

# **Case 8: Due Diligence on Retail Specialist**

Case Type: Market Entry  
Difficulty Level: Medium to Hard

# Due Diligence on Retail Specialist

## Problem Statement

A PE fund is thinking about acquiring a clothes retail specialist, a leader in the French market. The French clothing retail market is composed of 2 segments:

1. Urban: trendy, high quality, quite expensive
2. Suburban: mass market, lower quality, low prices

The PE fund hired us to help them assess whether this opportunity is worth bidding for.

## Additional Information (Provide Upon Request)

- Client has no specific financial target or timeline
- The target has total annual sales of \$800M across 800 stores. It consists of 4 brands on both the urban and suburban markets

# Due Diligence on Retail Specialist

## Sample Framework

### 1. Market

#### A. Industry

- Market size & growth
- French clothing retail trends
- Market segment characteristics

#### B. Market Drivers

- Buyer & supplier powers
- New entrants & substitutes
- Distribution channels

#### C. Regulations

### 2. Target

#### A. Quantitative

- Revenue models
- Key financials/specific assets
- Cost structure

#### B. Qualitative

- Competitive advantage
- Capabilities & resources
- Management structure/culture
- Expertise & knowledge

### 3. The Deal

#### A. Synergies

- Revenue synergies
- Cost synergies
- Operational synergies

#### B. Risks

- Implementation
- Regulation-trust
- External factors
- Management culture clash

### Business insights

- An ideal candidate should be able to come up with a framework that is appropriate for assessing a clothing retail company, and be able to ask relevant questions in each aspect to evaluate the target properly
- It's important to consider the synergies and risks of a deal even if the target is financially attractive. There could be potential problems down the road if the deal does not fit well with the PE fund

# Due Diligence on Retail Specialist

## Q1: Drivers for Price Drop

The market is flat in value over the last 5 years (~\$25Bn), but volumes have been growing over the same period (~2% per year). What could explain this?

## Sample Response

The only reason that can explain this situation is a decrease in price that offsets that increase in volume. Below are possible drivers of price reduction.

### 1. Customers

- Customers are looking for lower price and promotions
- Customers are buying more clothes with cheaper options

### 2. Competition

- New competitors in the market driving down the price
- More pricing comparison across competitors

### 3. Supply Chain

- Decrease in supply cost (outsourcing to lower cost countries)
- Decrease in raw material costs



# Due Diligence on Retail Specialist

## Q2: Store Profitability

An important metric in retail is floor space, specifically “sales/sq feet”. For one given brand, we have:

- 15 suburban stores - average size of 1600 sq ft and average profitability of \$1,500 per sq ft
- 20 urban stores - average size of 800 sq ft and average profitability of \$2,500 per sq ft

What is the total annual sale of this brand?

What is the average profitability (in terms of sales/sq feet) for the brand?

## Calculation

**Suburban:**  $1,600 * \$1,500 * 15 = \$36\text{M}$

**Urban:**  $800 * \$2,500 * 20 = \$44\text{M}$

**Total annual sale** = \$80M

**Average profitability** = Total sales / Total Size =  $\$80\text{M} / 40,000 \text{ sq ft} = \$2,000 / \text{sq ft}$

# Due Diligence on Retail Specialist

## Q3: Target's Investment Plan

The PE fund looked into the target company's investment plan over the last few years

|               |                                                                             |
|---------------|-----------------------------------------------------------------------------|
| <b>Year 1</b> | The brand opened its first store and gross margin was 10%                   |
| <b>Year 2</b> | Same store but gross margin increased to 25%                                |
| <b>Year 3</b> | Same store but gross margin remained flat to 25%. Manager decided to expand |
| <b>Year 4</b> | 1 additional store; total gross margin was at 20%                           |
| <b>Year 5</b> | 3 additional stores; total gross margin was at 18%                          |
| <b>Year 6</b> | 10 additional stores; total gross margin was at 16%                         |

Overall gross margin kept decreasing over the years and the PE fund is worried about that. Is the fund right to worry about it? Why or why not?

# Due Diligence on Retail Specialist

## Q3: Target's Investment Plan

Additional Information to be provided upon request:

1. All stores are similar
2. No cannibalization or coordination diminishing returns
3. Opening new stores impact mainly fixed costs

## Sample Answer

- There is a “ramp up” in the first few years, with increasing brand awareness and gross margin
- Brand reached full potential at an average gross margin of 25%
- Average margin will tend to decrease as the brand continue to expand will be a mix of matured stores and new stores
- Opening new stores is financially expensive and will impact average gross margin in the beginning as it goes through a ramp up period
- The fund should also look at individual store gross margin to ensure mature stores are maintaining healthy gross margins

# Due Diligence on Retail Specialist

## Q4: Additional Store Expansion

Based on your previous answer, if there were 3 additional stores in year 7 and 2 additional stores in year 8, what do you expect to happen to the overall gross margin?

## Sample Answer

- The total margin should increase as the expansion slows down and more existing stores would reach their mature state. The overall impact of opening new stores on total gross margin should decrease
- An ideal candidate should be able to draw on this question to identify the conclusion and implication. Candidate should state that if the target's total gross margin increased after year 7 and 8, then the fund should not worry as the decrease in total gross margin previous years were due to rapid expansion

# Due Diligence on Retail Specialist

## Recommendation

Please make a recommendation for our client.

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommendation</b> | <p>Yes, the client should continue to look into this deal</p> <ul style="list-style-type: none"><li>• Despite the total market value is flat, the business volume is increasing. The decrease in price may drive the less competitive players out</li><li>• During this time, the target seems to have maintained a healthy margin and the interim drop in margin was due to the rapid expansion</li><li>• It looks like there is still room for further expansion until the market become saturated</li></ul> |
| <b>Risks</b>          | <ul style="list-style-type: none"><li>• This is a competitive and rapidly changing market that the client still don't know very much about and does not have the expertise in</li><li>• The target company's margin while healthy, is still fairly low even when the new store reaches full potential</li></ul>                                                                                                                                                                                                |
| <b>Next Step</b>      | <ul style="list-style-type: none"><li>• Conduct a market competitive analysis to understand how other competitors are playing in the market</li><li>• Analyze target company's organizational structure and culture to evaluate management/portfolio fit</li></ul>                                                                                                                                                                                                                                             |

# **Case 9: Princeton-Plainsboro Hospital**

Case Type: Profitability  
Difficulty Level: Medium to Hard

## 9. Princeton-Plainsboro Hospital - Prompt

### Problem Statement

Princeton-Plainsboro Hospital is a large New Jersey single-site hospital serving a wide range of patients. The hospital's board is concerned because they have noticed a decline in the hospital's earnings from medical services even though the number of patients have remained static. The hospital has hired you to help them figure out what the problem is and come up with a strategy for increasing earnings.

### Additional Information (Provide Upon Request)

- The hospital is large and serves a wide variety of range of patients
- No other information is needed

## 9. Princeton-Plainsboro Hospital - Structure

### Sample Framework

| 1. Market                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2. PP Hospital (Company)                                                                                                                                                                                                                                                                                                                                                                  | 3. Financials                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4. Growth opportunities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A. Market Size</b> <ul style="list-style-type: none"><li>Market growth (global and local)</li><li>Demand (segmented by services or specializations)</li></ul> <b>B. Competitors</b> <ul style="list-style-type: none"><li>Number of other competitors</li><li>Services offered</li><li>Price per operation / profit margin</li></ul> <b>C. Trends</b> <ul style="list-style-type: none"><li>Private and public insurance</li><li>Pandemics, supply shortages, etc.</li></ul> | <b>A. Product</b> <ul style="list-style-type: none"><li>Types of operations and services</li></ul> <b>B. Customers</b> <ul style="list-style-type: none"><li>Patient segments (e.g., long-term vs short-term care, check-ups vs operations)</li></ul> <b>C. Marketing strategy</b> <ul style="list-style-type: none"><li>What kind of operations / services are we encouraging?</li></ul> | <b>A. Revenue</b> <ul style="list-style-type: none"><li>Price (avg price per operation or department)</li><li>Volume (number of patients, insurance vs cash mix, service mix)</li><li>Payment timelines (how long to receive payment - private vs public insurance, uninsured)</li></ul> <b>B. Cost</b> <ul style="list-style-type: none"><li>Variable (labor, materials, marketing etc.)</li><li>Fixed (rent, loans, equipment, utilities, malpractice insurance, etc.)</li></ul> | <b>A. Pricing strategy</b> <ul style="list-style-type: none"><li>Increasing the price and/or offer more flexible payment plans</li><li>Bundle services</li></ul> <b>B. Volume dynamics</b> <ul style="list-style-type: none"><li>Marketing towards patients who need more expensive services</li></ul> <b>D. Cost reduction</b> <ul style="list-style-type: none"><li>Improve efficiency and patient turnover</li><li>Move certain services online (after-visit check up).</li></ul> <b>D. Expansion</b> <ul style="list-style-type: none"><li>Expand organically, through partnership, or M&amp;A</li></ul> |

#### Interviewer guidance

- This case has two main components. First, investigate the decline in earnings for the hospital, then use that as a basis to develop a new corporate strategy.
- The investigation of earnings will be more structured with defined information and calculations to be completed. However, the strategy development is open-ended and gives the interviewee the opportunity to be creative.



## 9. Princeton-Plainsboro Hospital - Math Prompt

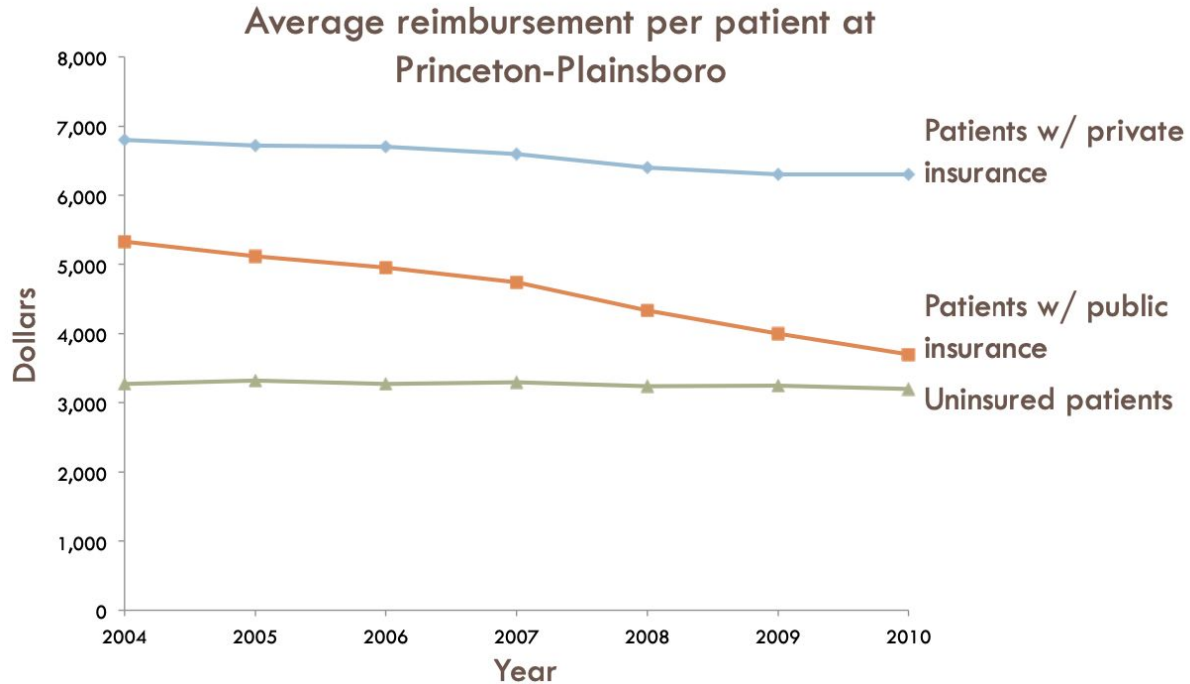
### Math prompt

Given the Exhibit 1, what was the operating margin for the hospital in 2004 and what is it in 2010?

### Additional Information (Provide Upon Request)

- Per patient costs:
  - 2004: \$4,000
  - 2006: \$4,400
  - 2008: \$4,600
  - 2010: \$4,800
- Patient Mix by Insurance type was constant over time:
  - Private: 50% - Public: 40% - Uninsured: 10%

## Exhibit 1 - Average Reimbursement per patient



## 9. Princeton-Plainsboro Hospital - Math Prompt

### Calculations

| Product | Calculations                                                                                                                                                                                               | Operating Margin % |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 2004    | 2004: Average per patient revenue for all patients =<br>$6,800 \times (50\%) + 5,300 \times (40\%) + 3,300 \times (10\%) = \$5,850$ .<br>Margin = (Rev-Costs)/Rev = $(\$5,850 - 4,000) / \$5,850 = 31.6\%$ | 31.6%              |
| 2010    | Average per patient revenue for all patients = $6,300 \times (50\%) + 3,700 \times (40\%) + 3,300 \times (10\%) = \$4,960$ .<br>Margin = (Rev-Costs)/Rev = $(\$4,960 - 4,800) / \$4,960 = 3.2\%$           | 3.2%               |

- Takeaways: overall, revenues are dropping
  - Operating margin has dropped ten fold from 32% to 3.2%
  - Biggest problem is drop in public insurance (Medicare/Medicaid) reimbursement amounts.
  - Private reimbursement also fell
  - Costs are rising steadily over time

## 9. Princeton-Plainsboro Hospital - Brainstorming

### Final Recommendation

Given the answer to the first question, what would you recommend the hospital do?

#### 1. Revenue

##### A. Price

- Charge more for services (could partner with other hospitals in state when negotiating reimbursement rates)

##### B. Marketing

- Attract more high margin private insurance patients via advertising, etc.

##### C. Expansion

- Acquire, joint venture, or do some sort of corporate development with local specialty clinic that serves private insurance patients with high reimbursement

##### B. Lobbying

- Lobby the government to raise public reimbursement rates

#### 2. Costs

##### A. Suppliers

- Partner with other hospitals in the state when negotiating prices

##### B. Customer targeting

- Target lower cost patients with selective advertising, etc. (e.g. target young rather than old)

##### C. Synergies through M&A

- Find efficiencies / synergies somewhere with a purchase of a clinic
- Outsource some services that can be handled more efficiently elsewhere (urgent care clinics).

#### Interviewer guidance

Make sure the candidate considers corporate development with a local specialty clinic that has desirable (profitable) patients

- Can acquire by purchasing
- Can enter into a joint venture or partner to provide some services in hospital, split profits with external specialist doctor
- Can try entering niche specialties alone by opening up a new office and hiring staff

## 9. Princeton-Plainsboro Hospital - Recommendation

### Recommendation

Please make a recommendation for our client.

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommendation</b> | <p>Our analysis observed that the operating margin has dropped ten fold from 32% to 3.2% and that the biggest problem is in the public insurance reimbursement amounts. In response, we recommend</p> <ul style="list-style-type: none"><li>● Partnering to other hospitals to renegotiate reimbursement rates and, in particular, to lobbying the government to raise the public rates.</li><li>● Adjust the marketing strategy to attract more high margin private insurance patients</li><li>● Consider acquiring a local speciality clinic that focus on high margin operations</li></ul> |
| <b>Risks</b>          | <ul style="list-style-type: none"><li>● Lobbying efforts could incite public backlash and would need to make a convincing case as to why the rates are to low</li><li>● Targeting higher margin activities could provoke a competitive response</li></ul>                                                                                                                                                                                                                                                                                                                                     |
| <b>Next Step</b>      | <ul style="list-style-type: none"><li>● Develop a marketing campaign to educate consumers as to the costs and values of the services in order to justify to the increase in reimbursement rates</li><li>● Perform an analysis of competitors current strategy and how they might respond</li></ul>                                                                                                                                                                                                                                                                                            |

# Case Math Drills

# Problem Set 1

1. You're considering a upfront \$5 million investment to switch to an alternative product B that will bring in additional \$2 millions in revenue but cost \$750K more than the current product A to produce. What is the payback period for this investment?
2. A competitor brand sells product for \$8 with a \$6 margin. Our product sell for \$12 and our unit cost is \$4. By what dollar value do we need to cut costs to have the same percentage profit margin as our competitors?
3. A new hybrid can travel 50% more MPG in comparison to the original car which travels 10 MPG. What is the annual cost saving if the owner on average drives 12,000 miles and fuel costs average \$3.00/gallon
4. A microwave costs \$100 to produce. The producer made a total profit of \$6 million by selling microwaves. If the markup percentage is 120%, how many microwaves did the producer sell?
5. At a company Christmas party, 1/2 in attendance are company employees. Assume that the rest of the attendees are spouses and friends. What percentage of the attendees are friends if there are 50% more spouses than friends?
6. If two planes leave the Philadelphia International Airport at 1:00 PM, how many miles apart will they be at 3:00 PM if one travels directly south at 150 mph and the other travels directly west at 200 mph?
7. A company with revenue of \$99 million is operating at a loss that is equivalent to 10% of its costs. By how much does cost need to decrease so that 10% of its costs are profits?

## Problem Set 2

1. AppleA is evaluating a project with an initial investment of \$35,000 and is expected to generate a revenues of \$10,000 in first year, \$17,000 in second year and \$19,000 in third year. The discount rate is expected to be 12%. Should Apple A invest in the project?
2. A Private Equity (PE) fund is considering a \$100M investment in a company expected to generate profits of \$10M in the first year. The fund typically invests keeping a return of 12% in target. Is this a feasible investment for the company?
3. A multiplex is planning to launch a line of casual clothing with a store in its complex. The expectation is that the store footfall should increase by \$200,000 annually. To estimate the actual footfall increase, the multiplex hired a market research company which generated the following results:

| Possible increase in attendance | Annual revenue (\$) | Probability |
|---------------------------------|---------------------|-------------|
| 3%                              | 135,000             | 20%         |
| 5%                              | 225,000             | 40%         |
| 7%                              | 315,000             | 30%         |
| 9%                              | 405,000             | 10%         |



## Problem Set 2

4. A burger company sells 30,000 pieces of burger per year with price of each being \$3. The company makes a margin of 50% on burgers and is planning to also stock donuts in its stores now. It expects to sell 50,000 donuts each at a price of \$2 and a 60% margin. It is expected that 10% lesser burgers will now be sold as customers might switch from consuming burgers to donuts. Should the burger company go ahead with selling donuts?
5. Our client is expecting to spent \$300,000 on internet marketing for their new album in CD format. The cost of manufacturing the CD is \$2/CD, distribution is \$3/CD and royalty to the band are expected to be \$3/CD. Each CD sells for \$12. How many CDs do our client need to sell to break even on its investment?

# Solutions: Problem Set 1

1. Profit:  $\$2\text{M (Revenue)} - \$750\text{K (Cost)} = \$1.25\text{ M}$   
Payback period:  $\$5\text{M}/\$1.25\text{M} = 4\text{ years}$
2. Competitor margin =  $\$6/\$8 = 0.75$   
Our new margin =  $0.75 * \$12 = \$9$   
Our new cost =  $\$12 - \$9 = \$3$   
We need to cut cost by \$1 to be competitive
3. Original fuel cost =  $12,000\text{ miles}/10\text{ MPG} * \$3 = \$3,600$   
New fuel cost =  $\$3,600 * (10/15) = \$2,400$   
Annual fuel savings =  $\$3,600 - \$2,400 = \$1,200$
4. Unit price =  $120\% * \$100 = \$120$   
Unit sold =  $\$6\text{ million} / \$120 = 50,000$
5. Employees is 50%  
Friends + Spouses is 50%  
Friends = 20%, Spouses = 30%
6. South:  $150 * 2 = 300\text{ miles}$   
West:  $200 * 2 = 400\text{ miles}$   
Use triangle hypotenuse (Pythagoras Theorem:  $\text{Hypotenuse}^2 = (\text{Side } 1)^2 + (\text{Side } 2)^2$ ): 500 miles
7. Profit: We know that Operating cost - 99 =  $10\% * (\text{Operating cost})$   
This means, Operating costs = 110  
New Profit:  $99 - \text{Operating costs} = 10\% * (\text{Operating cost})$   
New Operating costs = 90  
Operating cost reduction = 20

## Solutions: Problem Set 2

1. The decision on whether to invest or not depends on if investing in the project is the best use of capital currently. This can be done in 2 ways:
  - Is return on investment > opportunity cost of capital (also known as WACC - Weighted Average Cost of Capital)
  - Is project NPV (Net Present Value) positive

In this example, we do not have opportunity cost of capital, hence need to make NPV calculation

$$\text{NPV} = -(\text{Present value of cash OUTflows}) + \text{Present value of cash INflows}$$

$$\text{NPV} = -\$35,000 + (\$10,000/(1+12\%)) + (\$17,000/(1+12\%)^2) + (\$19,000/(1+12\%)^3) = \$8,977 > 0 \text{ hence,}$$
  
investment in project makes sense

2. To make this decision, the fund needs to calculate ROIC - Return on Invested Capital

$$\text{ROIC} = \text{Return} / \text{Invested Capital}$$

In this example,  $\text{ROIC} = \$10\text{M}/\$100\text{M} = 10\%$  which is less than the target 12%. Hence the fund should not invest in this company basis ROIC

3. To make this decision, we would need to calculate the expected value of the revenue increase. If expected value > target (\$200,000) investing in the new business line makes sense

Expected value = Sum of expected value of individual cases

Expected value =  $3\% * \$135,000 + 5\% * \$225,000 + 7\% * \$315,000 + 9\% * \$405,000 = \$252,000$  which is greater than the target \$200,000, hence investment can be considered

## Solutions: Problem Set 2

4. One product eating into other product's sales is called cannibalization rate, 10% in this example  
To make this decision, we should calculate the net impact on burger company's profitability  
Expected earnings by selling donuts =  $50,000 * \$2 * 60\% = \$60,000$   
Expected loss in burger sales due to cannibalization =  $10\% * 30,000 * \$3 * 50\% = \$45,000$   
Since  $\$60,000 > \$45,000$ , the burger company can consider stocking donuts
5. We need to calculate breakeven volume here which is the number of CDs you need to sell to get back your invested money  
Breakeven volume = Investment / (Profit per unit)  
Profit per unit (also known as Contribution Margin) =  $\$12 - (\$2 + \$3 + \$3) = \$4$   
Breakeven volume =  $\$100,000 / \$4 = 25,000$  CDs

# Brainstorm Drills

# Brainstorm Exercise

1. You are nearly finished with a case for a shaving company that sells its products to younger men. Profits are down due to lower sales: all products are type of men's razors sold in retail stores nationally. How would you go about improving sales?
2. You are helping a movie theater solve their declining profitability problem. After analyzing some financial data, you realize that the declining profitability is driven mainly out of dwindling revenue. What are some ways your client can improve revenue?
3. Your client is a PE firm looking to invest in an Airforce and Defense company that manufactures military grade helicopters. Based on the analysis so far, this seems like a good opportunity for your client, but the client wants to better understand additional growth potential of this target company. What are other potential avenues the Airforce and Defense company should consider to sell helicopters?

# Sample Answer #1

1. Launching New Products
  - a. Women's razors
  - b. Products targeting older men
  - c. Different sizes products (travel pack)
  - d. Shaving cream, soap, other complements
2. Channel beyond Retail Stores
  - a. E-commerce (Amazon)
  - b. Hotels
  - c. Specialty stores/ Salons
3. Marketing of current Products
  - a. Social media (influencers)
  - b. Free samples
  - c. Email campaigns
  - d. Billboards

## Sample Answer #2

### 1. Pricing

- a. Tier pricing for different movies and days
- b. Tier pricing for different seat options in the theater
- c. Offer monthly movie pass and loyalty program

### 2. Marketing/Partnership

- a. Secure marketing through movie production companies
- b. Partner with online ticket vendors (ex: Fandango)
- c. Partner with credit card companies (ex: opening night early access)

### 3. Other Revenue Streams

- a. Offer more food and beverage options
- b. Rent out movie theaters for events
- c. Sell movie merchandise in the theater
- d. Cultural and management alignment



# Sample Answer #3

1. Government
  - a. Other US departments: coastguards, homeland security
  - b. State governments: state police, fire department
  - c. Non-US: armed forces of overseas governments
2. Commercial
  - a. Oilfield Services
  - b. Leisure and Travel (private tours & transportation)
  - c. Media companies/News channels
  - d. Shipping packages (UPS, FedEx, Amazon deliveries)
3. Non-profit
  - a. Emergency medical services
  - b. Transport supplies for disaster relief

# **Behavioral Interview Guidelines**

# Behavioral Interviews

- **Showcase EQ:** cases test your IQ - fit is your chance to show EQ
- **Be Memorable:** answer all questions with a story/ example - show vs. tell. Stories are compelling way to articulate and demonstrate fit for firm. And let personality shine!
- **Delivery Matters:** be authentic, confident, structured and succinct. Demonstrate presence in front of a client
- **Firms really care:** consulting is a highly interpersonal business
- **You're in control (unlike cases):** Use fit questions as a source of confidence

**Use fit prep to boost a strength or shore up a weakness**

# Personal Stories Framework

The key is to be structured and try out different outlines to see what works for you!

**Headline:** Tell the interviewer where the story is going (~15%)

**Situation:** Set up the problem or conflict/ give background (~15%)

**Action:** What did you do? (~55%)

**Result (RII):** Highlight the outcome, the impact on the organization and what you learned (~15%)

## A STAR

- Answer
- Situation
- Task
- Action
- Result

## SCARF

- Situation
- Complication
- Action
- Result
- Future Lessons

The interviewer should understand the “so what?” before and after your story

## Sample Questions: Individual Contribution

1. Tell me about your proudest accomplishment
2. How would you describe your top 2 strengths and top 2 weaknesses?
3. Tell me about a time you set an ambitious goal, and what did you do to achieve it?
4. What is the biggest asset you can bring to your project team?
5. What makes you a good fit for consulting?

## Sample Questions Leadership/Entrepreneurial Drive

1. Tell me about a time when you made an unpopular decision
2. Tell me about a time when you managed a difficult team member or client
3. What is an example of a time when you led a team through a difficult situation
4. Tell me about a time when you had to provide someone negative feedback
5. Tell me about a time when you lead a team where there were a lot of differing opinions

## Sample Questions: Persuasion/Influence

1. Describe a time you had to convince someone who had an opposing viewpoint
2. Tell me about a time you had a disagreement with your manager
3. Tell me about the best presentation you have made
4. How do you convince others to change their viewpoint
5. Tell me about a time when you had to get buy-in from stakeholders

## Sample Questions: Analytics/Problem Solving

1. What is an example of a project that you worked on that was highly analytical?
2. Tell me about a time when you used data and analytics to solve a problem
3. Tell me about a time when you had to simplify technical data
4. Tell me about a time when you used data to drive a conclusion. What assumptions did you make?
5. Describe an example of when you had to work with complex data. How did you handle it?

## Sample Questions: Challenge/Failure

1. Tell me about a time when you had to overcome an obstacle
2. Tell me about a time when you were wrong and someone had to convince you
3. Tell me about a time when you had to navigate a difficult situation under time pressure
4. Tell me about a time you didn't succeed as a leader
5. Tell me about a time when you failed to meet a deadline or goal

## Sample Questions: Teamwork/Collaboration

1. Tell me about a time when you had to work with people from different backgrounds
2. Tell me about your role in a team
3. How have you dealt with differences in a team?
4. Tell me about a time when you had to work with a teammate whom you clashed with
5. Tell me about a time when you were in a difficult teamwork situation. How did you respond?